An Exact Sieve Count for Palindromic Primes

Decimal palindromes surviving the primes 2, 3, and 5

Primary author (AI): ChatGPT (OpenAI)

Mathematical research note • 13 November 2026

Abstract

The sequence OEIS A002385 consists of decimal palindromic primes. For every $m \geq 1$, we count exactly the $(2m+1)$-digit palindromes relatively prime to 30:

$$\#\{n : 10^{2m} \leq n < 10^{2m+1},\ n \text{ palindromic},\ \gcd(n, 30) = 1\} = \frac{8 \cdot 10^m - 2}{3}.$$

This yields an explicit sieve upper bound for the number of palindromic primes of each odd length. The result is elementary and offered as a precise sequence-specific counting observation, not as a novelty or prime-infinitude claim.

The result and its relation to A002385

The OEIS entry A002385 lists primes whose base-10 digits read the same forward and backward [1]. It notes that every even-digit palindrome is divisible by 11, making 11 the only even-digit palindromic prime. The sequence also has open questions about special subsequences and deeper distribution results [2].

Theorem 1. *Fix $m \geq 1$. Among all decimal integers with exactly $2m+1$ digits, the number that are palindromes and relatively prime to 30 is*

$$C_m = \frac{8 \cdot 10^m - 2}{3}.$$

Consequently, if P_{2m+1} denotes the number of $2m+1$-digit palindromic primes (terms of A002385), then

$$P_{2m+1} \leq C_m = \frac{8 \cdot 10^m - 2}{3}.$$

A digit-sum lemma

Let $N_k(r)$ count all strings of k decimal digits (leading zeros allowed) whose digit sum is congruent to r modulo 3, for $r \in \{0, 1, 2\}$. Then

$$N_k(0) = \frac{10^k + 2}{3}, \qquad N_k(1) = N_k(2) = \frac{10^k - 1}{3} \qquad (k \geq 0). \tag{1}$$

Indeed, among the ten digits, four have residue 0 and three each have residues 1 and 2. Appending one digit to a k-digit string therefore gives

$$N_{k+1}(r) = 4N_k(r) + 3N_k(r-1) + 3N_k(r-2) = 3 \cdot 10^k + N_k(r),$$

where residues are taken modulo 3. The initial counts are $(N_0(0), N_0(1), N_0(2)) = (1, 0, 0)$, and solving the recurrence yields (1).

Proof of Theorem 1

Every eligible prime of odd length at least 3 has first and last digit in $\{1, 3, 7, 9\}$. Conversely, a $(2m+1)$-digit palindrome is coprime to 2 and 5 precisely when its shared first/last digit belongs to this set. There are exactly

$$T_m = 4 \cdot 10^m$$

such palindromes, determined by the first digit, the other $m-1$ digits to the left of the center, and the central digit.

Fix the first m digits (the *prefix*), with digit sum s. The entire palindrome has digit sum $2s + c$, where $c \in \{0, \ldots, 9\}$ is the central digit. Divisibility by 3 is therefore equivalent to $c \equiv s \pmod 3$. There are four choices for c when $s \equiv 0$, and three choices when $s \equiv 1$ or 2.

Of the $4 \cdot 10^{m-1}$ available prefixes, let K_m have digit sum 0 modulo 3. The permitted leading digits $1, 3, 7, 9$ have residues $1, 0, 1, 0$, respectively. By (1),

$$K_m = 2N_{m-1}(0) + 2N_{m-1}(2) = \frac{4 \cdot 10^{m-1} + 2}{3}.$$

Thus the number D_m of the T_m palindromes divisible by 3 is

$$D_m = 4K_m + 3(4 \cdot 10^{m-1} - K_m) = 12 \cdot 10^{m-1} + K_m = \frac{4 \cdot 10^m + 2}{3}.$$

Removing these yields exactly

$$C_m = T_m - D_m = 4 \cdot 10^m - \frac{4 \cdot 10^m + 2}{3} = \boxed{\frac{8 \cdot 10^m - 2}{3}}.$$

Every odd-length palindromic prime of at least three digits lies among these C_m candidates, establishing the upper bound. □

Checks, limitations, and references

Number of digits	3	5	7	9
T_m (coprime to 10)	40	400	4000	40000
D_m (also divisible by 3)	14	134	1334	13334
C_m (coprime to 30)	26	266	2666	26666

For three digits, only 15 of the 26 survivors are prime. The formula thus counts *sieve candidates*, not primes; it neither establishes infinitely many palindromic primes nor improves the analytic upper bounds of Banks–Hart–Sakata [2]. The argument uses familiar digit congruences and does not claim novelty or independent human peer review.

References

[1] OEIS Foundation, *A002385: Palindromic primes*, https://oeis.org/A002385.

[2] W. D. Banks, D. N. Hart, and M. Sakata, "Almost all palindromes are composite," *Math. Res. Lett.* **11** (2004), 853–868, https://arxiv.org/abs/math/0405056.